

TESTS FOR CONVERGENCE/DIVERGENCE

Lesson 28

Given an infinite series $\sum_{n=1}^{\infty} u_n$, we can apply tests

to determine whether they are convergent or divergent. The first test to apply is the divergence test.

Test 1: (Divergence test)

An infinite series $\sum_{n=1}^{\infty} u_n$ cannot be convergent unless

$$\lim_{n \rightarrow \infty} u_n = 0.$$

* If $\lim_{n \rightarrow \infty} u_n \neq 0$, then $\sum_{n=1}^{\infty} u_n$ is divergent.

Example 1

Use divergence test to determine convergence/divergence of the following:

(a) $\sum_{n=1}^{\infty} 8\left(\frac{3}{4}\right)^n$

(b) $\sum_{n=1}^{\infty} \frac{n}{n+2}$

(c) $\sum_{n=1}^{\infty} \frac{1}{n}$

(d) $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$

Soln

(a) $\lim_{n \rightarrow \infty} 8\left(\frac{3}{4}\right)^n = 0 \therefore$ Is convergent (Indeed G.P $|r| = \frac{3}{4} < 1$.)

(b) $\lim_{n \rightarrow \infty} \frac{n}{n+2} = \lim_{n \rightarrow \infty} \frac{1}{1+\frac{2}{n}} = 1 \neq 0 \therefore$ Divergent by test 1.

(c) $\lim_{n \rightarrow \infty} \frac{1}{n} = 0 \therefore$ ~~Convergent~~ by test 1 (Wrong) $\left\langle \begin{array}{l} \text{We know} \\ \text{Harmonic series is } \underline{\text{divergent}} \end{array} \right\rangle$

(d) $\lim_{n \rightarrow \infty} \frac{1}{n(n+1)} = 0 \therefore$ Convergent $\left\langle \begin{array}{l} \text{We know} \\ \text{Telescopic series is } \underline{\text{convergent}} \end{array} \right\rangle$

Note: Test 1 - divergence test is not reliable to test convergence

Eg: $\sum_{n=1}^{\infty} \frac{1}{n}$, $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ shows convergence but are divergent.

Test 2: RATIO TEST

Ratio test (or D'Alembert ratio test) is used to test for convergence/divergence of infinite series with positive terms

Given $\sum_{n=1}^{\infty} U_n$ where U_n - Positive term sequence.

RATIO TEST (states)

* If $\lim_{n \rightarrow \infty} \frac{U_{n+1}}{U_n} < 1$, then $\sum_{n=1}^{\infty} U_n$ is Convergent.

* If $\lim_{n \rightarrow \infty} \frac{U_{n+1}}{U_n} > 1$, then $\sum_{n=1}^{\infty} U_n$ is divergent

* If $\lim_{n \rightarrow \infty} \frac{U_{n+1}}{U_n} = 1$, then ratio test is inconclusive
(ie ratio test fails, try another test)

Example 2

Use ratio test to determine whether the following series are Convergent or divergent.

$$(a) \sum_{n=1}^{\infty} \frac{2n-1}{2^n}$$

$$(b) \sum_{n=1}^{\infty} \frac{3^{n-1}}{4n+1}$$

$$(c) \sum_{n=1}^{\infty} \frac{2n^2+1}{3n^2+2}$$

Solution

$$(a) \text{ Here } U_n = \frac{2n-1}{2^n} ; U_{n+1} = \frac{2(n+1)-1}{2^{n+1}} = \frac{2n+1}{2^{n+1}}$$

$$\text{Then } \frac{U_{n+1}}{U_n} = \frac{2n+1}{2^{n+1}} \cdot \frac{2^n}{2n-1} = \frac{1}{2} \left(\frac{2n+1}{2n-1} \right) = \frac{1}{2} \left(\frac{2 + \frac{1}{n}}{2 - \frac{1}{n}} \right)$$

$$\lim_{n \rightarrow \infty} \frac{U_{n+1}}{U_n} = \lim_{n \rightarrow \infty} \frac{1}{2} \left(\frac{2 + \frac{1}{n}}{2 - \frac{1}{n}} \right) = \frac{1}{2} \left(\frac{2+0}{2-0} \right) = \frac{1}{2} < 1$$

Hence $\sum_{n=1}^{\infty} \frac{2n-1}{2^n}$ is Convergent by ratio test.

(b) Here $U_n = \frac{3^{n-1}}{4n+1}$; $U_{n+1} = \frac{3^{(n+1)-1}}{4(n+1)+1} = \frac{3^n}{4n+5}$

$$\frac{U_{n+1}}{U_n} = \frac{3^n}{4n+5} \cdot \frac{4n+1}{3^{n-1}} = 3 \left(\frac{4n+1}{4n+5} \right) = 3 \left(\frac{4 + \frac{1}{n}}{4 + \frac{5}{n}} \right)$$

$$\lim_{n \rightarrow \infty} \frac{U_{n+1}}{U_n} = \lim_{n \rightarrow \infty} 3 \left(\frac{4 + \frac{1}{n}}{4 + \frac{5}{n}} \right) = 3 \left(\frac{4+0}{4+0} \right) = 3 > 1$$

$\therefore \sum_{n=1}^{\infty} \frac{3^{n-1}}{4n+1}$ is divergent by ratio test

(c) Here $U_n = \frac{2n^2+1}{3n^2+2}$; $U_{n+1} = \frac{2(n+1)^2+1}{3(n+1)^2+2} = \frac{2n^2+4n+3}{3n^2+6n+5}$

$$\frac{U_{n+1}}{U_n} = \frac{2n^2+4n+3}{3n^2+6n+5} \cdot \frac{3n^2+2}{2n^2+1} = \left(\frac{3n^2+2}{3n^2+6n+5} \right) \left(\frac{2n^2+4n+3}{2n^2+1} \right) = \left(\frac{3 + \frac{2}{n^2}}{3 + \frac{6}{n} + \frac{5}{n^2}} \right) \left(\frac{2 + \frac{4}{n} + \frac{3}{n^2}}{2 + \frac{1}{n^2}} \right)$$

$$\lim_{n \rightarrow \infty} \frac{U_{n+1}}{U_n} = \lim_{n \rightarrow \infty} \left(\frac{3 + \frac{2}{n^2}}{3 + \frac{6}{n} + \frac{5}{n^2}} \right) \left(\frac{2 + \frac{4}{n} + \frac{3}{n^2}}{2 + \frac{1}{n^2}} \right) = \left(\frac{3+0}{3+0+0} \right) \left(\frac{2+0+0}{2+0} \right) = 1$$

Since $\lim_{n \rightarrow \infty} \frac{U_{n+1}}{U_n} = 1$, Ratio test is inconclusive

i.e Ratio test fails try another test,

Using divergent test on $\sum_{n=1}^{\infty} \frac{2n^2+1}{3n^2+2}$

$$\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} \frac{2n^2+1}{3n^2+2} = \lim_{n \rightarrow \infty} \left(\frac{2 + \frac{1}{n^2}}{3 + \frac{2}{n^2}} \right) = \frac{2}{3} \neq 0$$

Hence $\sum_{n=1}^{\infty} \frac{2n^2+1}{3n^2+2}$ is divergent (by divergent test).